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7.3.3 Convergence of splitting methods

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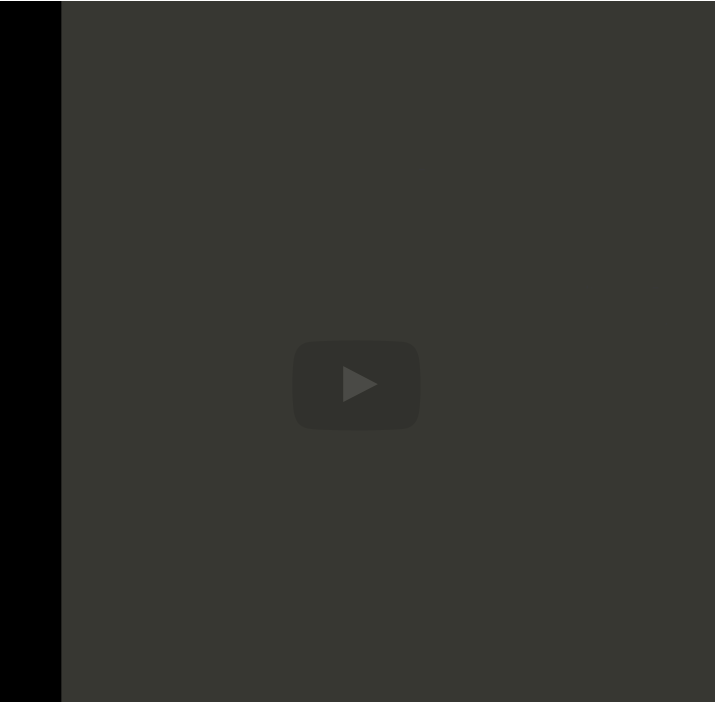
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“ ”

each other, and what you get is that M times the quantity $x^{(k+1)}$ minus the actual solution is equal to N times $x^{(k)}$ minus the actual solution. Okay? And this is good because now we're talking about how close is $x^{(k+1)}$ to x relative to how close $x^{(k)}$ was to x . And then it actually becomes beneficial to bring M to the other side. So we get that $x^{(k+1)}$ minus x is equal to M inverse times N times $x^{(k)}$ minus x . Ah, and then we can

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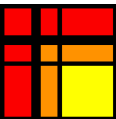
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Reading assignment

1/1 point (graded)



Read Unit 7.3.3 of the notes. [[LINK](#)]

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Homework 7.3.3.1

1/1 point (graded)
Why are the choices of M and N used by the Jacobi iteration and Gauss-Seidel iteration convenient?

(Mark all correct answers.)

- ☒ Solving with M is convenient and cheap.
- ☒ The sparsity in A is preserved in M and N



Solution:

For a complete solution see the [notes](#).

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Answers are displayed within the problem

Homework 7.3.3.2

1/1 point (graded)
Let $A = M - N$ be a splitting of matrix A . Let $x^{(k+1)} = M^{-1} (Nx^{(k)} + y)$. Show that

$$x^{(k+1)} = x^{(k)} + M^{-1}r^{(k)}, \text{ where } r^{(k)} = y - Ax^{(k)}.$$

- ☒ done



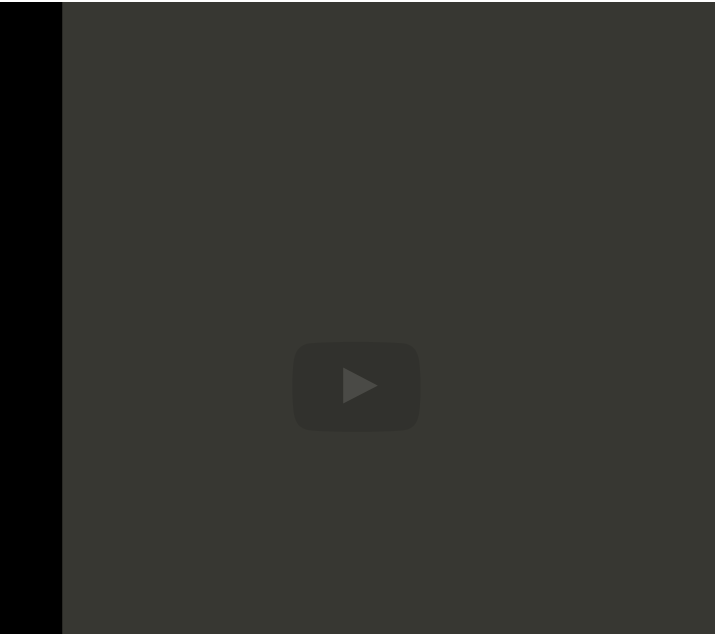
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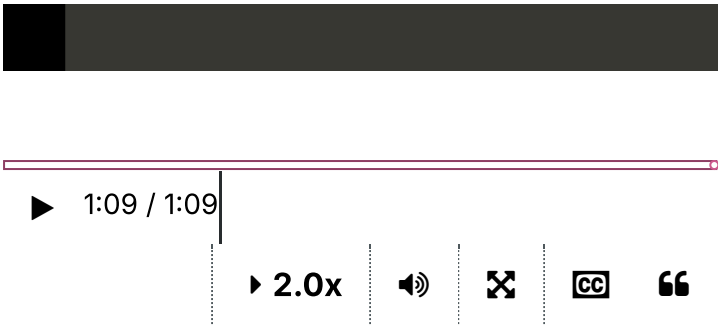
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property that for some norm $\|M^{-1}N\|$ is less than one requires us to learn a little bit about eigenvalues. And the spectral radius of the matrix comes in. And that's a future topic in this course. So we're going to defer on that. However, you can sorta contemplate under what circumstances is $\|M^{-1}N\|$ in norm



inverse with norm
small. Okay? What is
the best way of
splitting A into M and N
such that you
converge in the

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Homework 7.3.3.5

1/1 point (graded)
Given nonsingular $A \in \mathbb{R}^{n \times n}$, what splitting $A = M - N$ will give the fastest convergence to the solution of $Ax = y$?

Split $A = L + D + U$, where L , D , and U equal the strictly lower triangular, diagonal, and strictly upper triangular parts of A .

- ☐ $M = D$
- ☐ $M = L$
- ☒ $M = A$



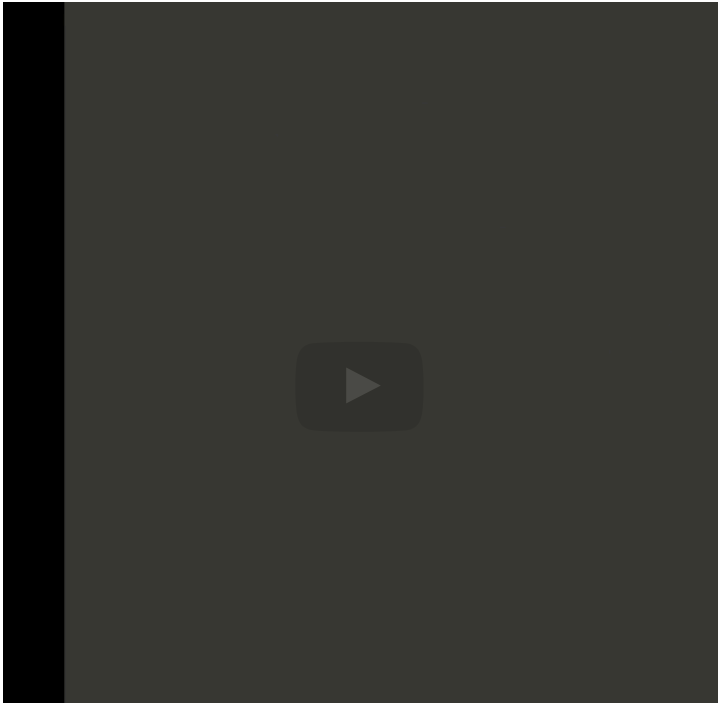
Solution:

For a complete solution see the [notes](#).

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lower triangular
system is cheap. So
that's convenient.
And then we saw that
you know even
though the splitting
matrix for the
symmetric Gauss-
Seidel is
quite a bit more
complicated, in
practice you can
think of it as solving
with two
sparse triangular
matrices. So that's
convenient as well.
So there's this
delicate balance
between picking an
M and N such that

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